

Day 12

Handmade

10 20 30

camlin

Date

## Linked List

101

Nothing to write particularly but  
lets code.

$2 \times 2 = 4$

```
public class Day16 {  
    static class Node {  
        int data;  
        Node next;
```

```
        Node (int data) {  
            this.data = data;  
            this.next = null;
```

```
        }  
    }  
}
```

```
static class LinkedList {  
    Node head;
```

```
    void addFirst (int data) {
```

```
        Node newNode = new Node (data);
```

```
        if (head == null) {
```

```
            head = newNode;
```

```
        }  
        return;
```

```
    }  
}
```

```
newNode.next = head;
```

```
head = newNode;
```

```
    }  
}
```

For  
add first

P.T.O.



```

void printList() {
    Node currNode = head;
    while (currNode != null) {
        cout << currNode->data << " ";
        currNode = currNode->next;
    }
}

```

For  
printing  
linked list.

```

void addLast (int data) {
    if (head == null) {
        addFirst (data);
        return;
    }
    Node newNode = new Node (data);
    Node currNode = head;
    while (currNode->next != null) {
        currNode->next = newNode;
    }
}

```

For  
adding  
number  
at last

```

int size () {
    int count = 0;
    Node currNode = head;
    while (currNode != null) {
        count++;
        currNode = currNode->next;
    }
    return count;
}

```

For finding  
size



```

void add (int index, int data) {
    if (index == 0) {
        addFirst(data);
        return;
    }
    Node currNode = head;
    Node newNode = new Node(data);
    for (int i = 0; i < index - 1; i++) {
        currNode = currNode.next;
    }
    newNode.next = currNode.next;
    currNode.next = newNode;
}

```

For  
adding  
a  
node  
between

```

void removeFirst() {
    if (head == null) {
        return;
    }
    head = head.next;
}

```

To remove  
first

```

void removeLast() {
    if (head == null) {
        return;
    }
    if (head.next == null) {
        head = null;
        return;
    }
    Node currNode = head;
    while (currNode.next.next != null) {
        currNode = currNode.next;
    }
    currNode.next = null;
}

```

To remove  
last